WASHINGTON MBELEKI

SOFTWARE ENGINEER | DATA ENGINEER

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PROFILE

Software engineer targeting the Data and AI Engineer position at WatersEdge Solutions. Brings practical experience in building backend services, API integrations, and relational persistence layers. Direct technical alignment for this role includes ai, api, apis, data, database, engineering. Proven track record in relational persistence, query optimization, ETL concepts, and data-backed backend services, supported by award-winning hackathon achievements and academic peer tutoring at WeThinkCode_.

DATA ENGINEERING SKILLS

Languages & Query	Python, SQL, Java, JavaScript, TypeScript, Bash/Shell
Data Engineering & Storage	SQLite, PostgreSQL, Cloudflare D1 (Serverless Relational), Cloudflare KV (Key-Value Caching), Cloudflare R2 (Object Storage), Relational Database Design (RDBMS), ETL Pipeline Construction, Data Ingestion & Transformation, Data Validation & Quality Checks, TF-IDF & Cosine Similarity Matching
Backend, Web & APIs	REST APIs, Flask, Node.js, Hono, Cloudflare Workers, Microservices, JWT Authentication, Third-Party API Integration (MTN MoMo, Google Calendar), Socket & Network Programming, n8n Workflow Automation
Workflow, AI & Tooling	Linux/Ubuntu, Docker, Git/GitHub, GitLab CI/CD Automation, Test-Driven Development (TDD), JUnit 5, Maven, Generative AI & LLM API Integration, Prompt Engineering & Automated Test Generation, Web Scraping (BeautifulSoup), Agile Software Development

EXPERIENCE

Peer Tutor & Bootcamp Assistant | *WeThinkCode_* Oct 2025 - Present

- Facilitate hands-on sessions covering Python fundamentals, Java object-oriented design, and Git workflows for incoming cohorts.
- Conduct code reviews, help troubleshoot systems, and reinforce test-driven development (TDD) and maintainable coding practices.

SELECTED PROJECTS

Automated Job Pipeline & Tailored PDF Resume Engine | *Personal Project*

- Engineered an automated lead pipeline scraping public job feeds using Python, BeautifulSoup, and backoff retry handling.
- Implemented TF-IDF cosine similarity scoring to match candidate profiles against incoming job descriptions.
- Built a Jinja2 & xhtml2pdf rendering engine to automatically generate job-tailored PDF CVs and managed workflow tracking via a Streamlit dashboard and SQLite.

MoCredit - AI Credit Scoring & Micro-Lending Mini App | *2nd Place, MTN MoMo FinTech Hackathon, 2026*

- Worked on a fintech application using a Node.js/Hono backend on Cloudflare Workers, Cloudflare D1 for persistent storage, and Cloudflare KV for score caching.
- Integrated MTN MoMo Collections (RequestToPay) and Disbursements APIs, with an n8n-orchestrated AI coaching workflow.

Thuso Health - Digital Health Passport & Smart Clinic Finder | *Top 3, G13 Health Challenge*

- Co-built an offline-first health passport and clinic finder with backend APIs using Hono/Cloudflare Workers, D1 records, R2 medical attachments, and JWT authentication.
- Developed data-informed clinic recommendations using travel-time, queue-congestion, and load-shedding/solar-power information; supported offline caching and sync-on-reconnect.

TSUMO Network - Backend Data Architecture

- Co-engineered Python-based REST API endpoints handling media-streaming metadata workflows and user request routes.
- Implemented server-side business logic to automate complex, usage-based revenue allocation and payout calculations.
- Integrated application layers with persistent relational database storage to ensure safe, validated transaction logging.

Code Clinic Booking System | *WeThinkCode_ course project*

- Built a booking application for scheduling and managing academic code clinic sessions using Python, Flask, SQLite, and Google Calendar API integrations.

Robot World Simulation Platform | *WeThinkCode_ course project*

- Built a Java multiplayer client-server simulation using Maven, object-oriented design, test-driven development, Swing, and socket/network programming; refactored code in response to complexity warnings.

EDUCATION

Occupational Diploma in Software Engineering | WeThinkCode_ Oct 2025 - Expected Feb 2027

Johannesburg, South Africa | *Relevant coursework:* Software Engineering Principles, Data Structures & Algorithms, Object-Oriented Design, Concurrent Systems, CI/CD, and Database Management.

TRAINING & CERTIFICATIONS

- Data Engineering Basics for Everyone — edX, Aug 2026.

- Relational Database Management Systems — IBM, Sep 2026.
- Generative AI for Software Engineering — WeThinkCode_; prompt engineering, AI-assisted refactoring, test-suite generation, and LLM API integrations.

ROLE-FIT SUMMARY

> *Role-Fit Summary (Data and AI Engineer @ WatersEdge Solutions): Key matching strengths include practical competence in ai, api, apis, data, database, engineering, test-driven development (TDD), and relational persistence. Demonstrates immediate engineering value for WatersEdge Solutions's development goals with an ongoing commitment to mastering scalable data pipeline architectures.*

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Strategic Brief: Candidate Positioning & Value Proposition

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POSITIONING

Software engineer targeting entry-level data engineering roles, with hands-on project evidence across relational storage, API integration, caching, and data-informed application features. MoCredit and Thuso Health show how I connect data and services to support application workflows. My next step is to extend this foundation into clearly demonstrated ingestion, transformation, data-quality checks, and repeatable pipeline operation.

TARGET ROLES & WORKING SETUP

- **Roles:** Entry-Level / Junior Data Engineer; Data Engineering Intern or Trainee; Graduate Data Engineer; Junior ETL Developer; backend roles with substantial SQL and data-processing work.
- **Environment:** A team with mentorship, code review, clear data-quality expectations, and opportunities to build and operate small pipelines.
- **Application Strategy:** Prioritize roles that value Python, SQL, backend fundamentals, and learning potential. Treat Spark, Airflow, dbt, and cloud data warehouses as learning targets unless separate hands-on evidence is available.

VALUE PROPOSITION

- **Data-Backed Application Workflows:** MoCredit combines D1 persistence, KV caching, financial API integrations, and workflow automation.
- **Operational Data Features:** Thuso Health recommendations use travel-time, queue-congestion, and energy-availability information, alongside offline caching and sync considerations.
- **Engineering & Backend Discipline:** TSUMO Network, Code Clinic Booking, and peer tutoring provide evidence of API/service work, Python/SQLite, debugging, testing, code review, and teamwork.

PROOF POINTS

Project	What It Demonstrates	Stack
MoCredit	Persistent storage, caching, external API integration, workflow automation; 2nd place at MTN MoMo FinTech Hackathon 2026.	`Node.js, Hono, Workers, D1, KV, n8n`
Thuso Health	Data-informed recommendations, offline-first sync, R2 medical storage, JWT auth.	`Hono, Workers, D1, R2, JWT`
Job Pipeline Engine	Web scraping, TF-IDF semantic matching, SQLite tracking, and dynamic PDF generation.	`Python, BeautifulSoup, TF-IDF, SQLite, Streamlit, xhtml2pdf`
TSUMO Network	Backend REST APIs, media streaming metadata, automated revenue payout logic.	`Python, REST APIs, Relational Persistence`
Code Clinic	Database-backed booking workflow and Google Calendar integration.	`Python, Flask, SQLite, Google Calendar API`

STRATEGIC BRIEF: GAPS & DEVELOPMENT PLAN

KNOWN GAPS & HOW I AM CLOSING THEM

- **Standalone PySpark distributed data processing:** Build a large-scale data ingestion and transformation pipeline utilizing PySpark and SQL window functions for batch analytics.
- **Advanced SQL depth & optimization:** Add explicit public repo demonstrations featuring complex CTEs, window functions, indexing strategies, data deduplication, and execution plan optimization.
- **Data orchestration & scheduling:** Integrate Apache Airflow or Prefect to manage DAG schedules, retry policies, and logging for automated ETL runs.

FIRST 90 DAYS (PROPOSED)

Period	Focus	Evidence to Produce
Days 1-30	Build a complete Python + SQL ETL pipeline	Public API ingestion, validation/transformation, relational database loading, unit tests, and automated data quality checks.
Days 31-60	Improve query performance and pipeline orchestration	Window functions, indexing optimizations, automated retry/logging policies, and repeatable pipeline runs.
Days 61-90	Polish repositories and deliver tailored applications	Comprehensive architecture diagrams, project documentation, technical walkthroughs, and active candidate positioning.